

COURSE DESCRIPTION FORM

INSTITUTION FAST - National University of Computers and Emerging Sciences

PROGRAM (S) TO BE EVALUATED BS - Computer Science

A. Course Description

Course Title	Design and Analysis of Algorithms																		
Course Code	CS2009	Credit Hours	3 + 0																
Prerequisites by Course(s)	CS2001	Semester	Fall 2025																
Assessment Instruments (With tentative weights)	Semester Work 10% (5 assignments) Project 10% (one project) Midterm 30% (Midterm1:12.5, Midterm2: 17.5) Final 50% (Comprehensive end of semester exam)																		
Current Catalog Description	Main objective of course is to understand required skills and knowledge to design and analyze algorithms. Students will learn several algorithm design techniques and use of mathematical tools for empirical analysis of algorithms. Additionally, students will be able to learn effective problem-solving skills in computing through this course.																		
Textbook (or Laboratory Manual for Laboratory Courses)	1. Thomas H. Cormen et al, Introduction to Algorithms, 4 th Edition																		
Reference Material	1. Anany Levitin, Introduction to the design and analysis of algorithms, 3 rd Edition																		
Course Goals	A. Course Learning Outcomes (CLOs) <table border="1"> <thead> <tr> <th>CLO No.</th><th>Course Learning Outcomes</th><th>Bloom Taxonomy</th><th>Tools</th></tr> </thead> <tbody> <tr> <td>CLO-1</td><td>To apply acquired knowledge to solve computing problems complexities and proofs</td><td>C3 (Cognitive)</td><td>A1, A2, A3, A4, A5, M1, M2, F</td></tr> <tr> <td>CLO-2</td><td>To analyze complexities of different algorithms using asymptotic notations, complexity classes and standard complexity function</td><td>C4 (Cognitive)</td><td>A1, A2, A3, A4, A5, M1, M2, F</td></tr> <tr> <td>CLO-3</td><td>To evaluate generic algorithmic solutions</td><td>C5 (Cognitive)</td><td>A3, A4, A5, M1, M2, F</td></tr> </tbody> </table>			CLO No.	Course Learning Outcomes	Bloom Taxonomy	Tools	CLO-1	To apply acquired knowledge to solve computing problems complexities and proofs	C3 (Cognitive)	A1, A2, A3, A4, A5, M1, M2, F	CLO-2	To analyze complexities of different algorithms using asymptotic notations, complexity classes and standard complexity function	C4 (Cognitive)	A1, A2, A3, A4, A5, M1, M2, F	CLO-3	To evaluate generic algorithmic solutions	C5 (Cognitive)	A3, A4, A5, M1, M2, F
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[illegible]

		1	2	3	4	5	6	7	8	9	10
CLOs		✓	✓			✓					✓
4											
Topics Course (	Topics	No. of Weeks	Contact Hours	CLO							
	Basics of Algorithms, Mathematical Foundation, Growth of Function, asymptotic Notations. (Assignment 1 will be given at the start of week-2 and Project will be announced)	2	6	CLO-1							
	Divide and Conquer, Substitution Method, Recurrence-Tree Method, Master's Method.	2	6	CLO-2							
	Data Structures Review (Stack, Queue, Linked List, Hash Table, Binary Tree). Sorting (Merge, Insertion, Quick, Heap, Counting, Radix, Bucket) (Assignment 2 will be given)	1	3	CLO-3							
	Mid-I Examination	0.5	1	CLO-1,2,3							
	Geometric Algorithms (Introduction, Graham Scan, Close Points). String Matching	2	6	CLO-2,3							
	Dynamic Programming (Assignment 3 will be given)	2	6	CLO-3							
	Mid-II Examination	0.5	2	CLO-1,2,3,4							
	Greedy Algorithms, Graph Theory (Graph Categorization, Graph Terminology, Representation of Graphs, BFS & DFS, Strongly Connected Components, Greedy Algorithms: Kruskal's Algorithm, Prim's Algorithms, Bellman-Ford Algorithms, Dijkstra's Algorithm)	3	9	CLO-3							



National



	Total1751(Including 3 hours final exam)			
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	8	20	20	0
Oral and Written Communications	Every student is required to submit all 5 assignments and potential oral project presentations.			

Instructor Name __Dr. Nasir Uddin__

Instructor Signature _____

Date __13-08-2025__